



Coordinated Multi-Robot Navigation with Formation Adaptation

Zihao Deng¹, Peng Gao², Williard Joshua Jose¹, Christopher Reardon³, Maggie Wigness⁴, John Rogers⁴, and Hao Zhang¹

¹ Human-Centered Robotics Lab, UMass Amherst, USA

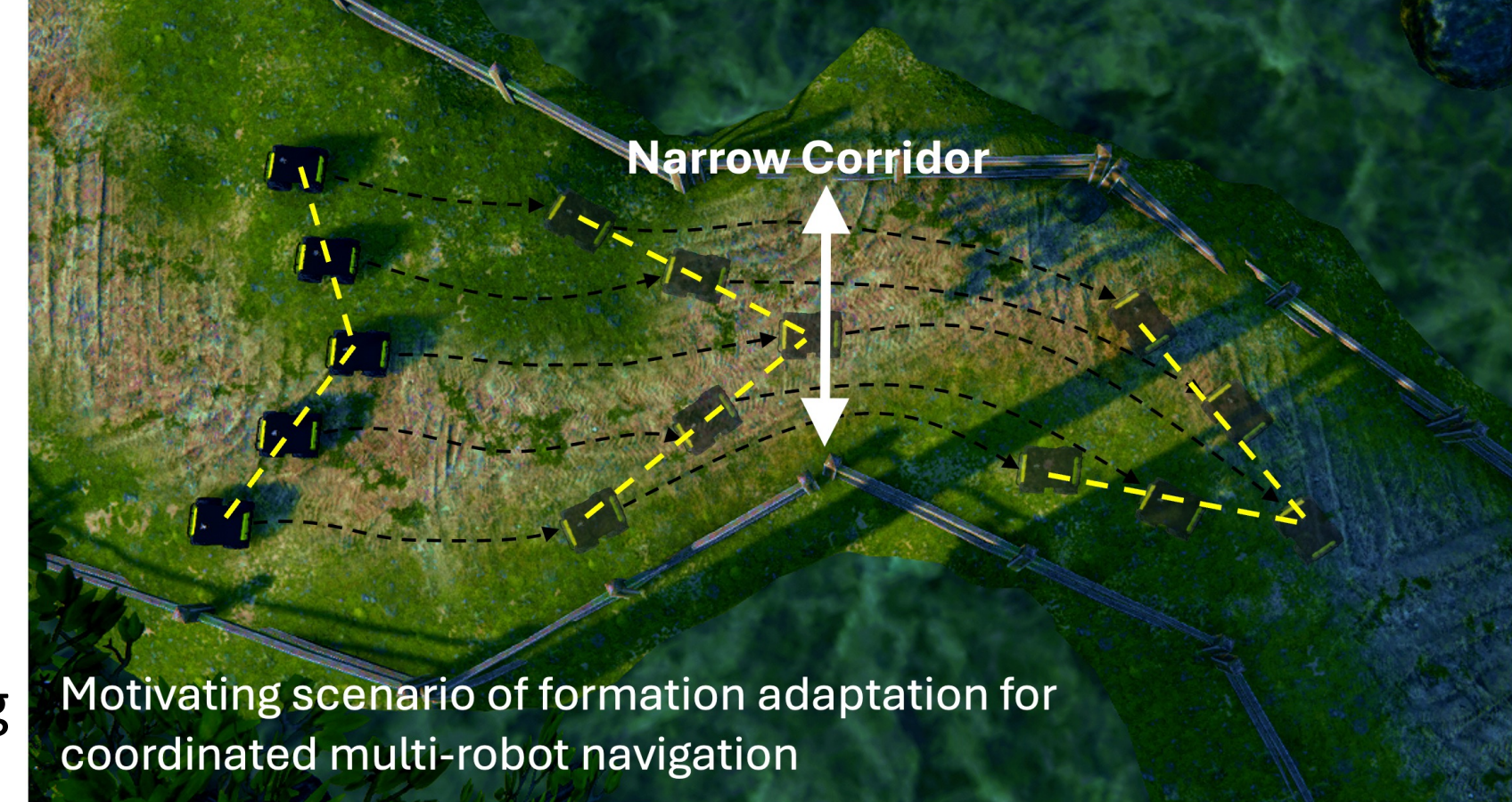
² North Carolina State University, USA

³ University of Denver, USA

⁴ US DEVCOM Army Research Laboratory (ARL), USA

Motivation

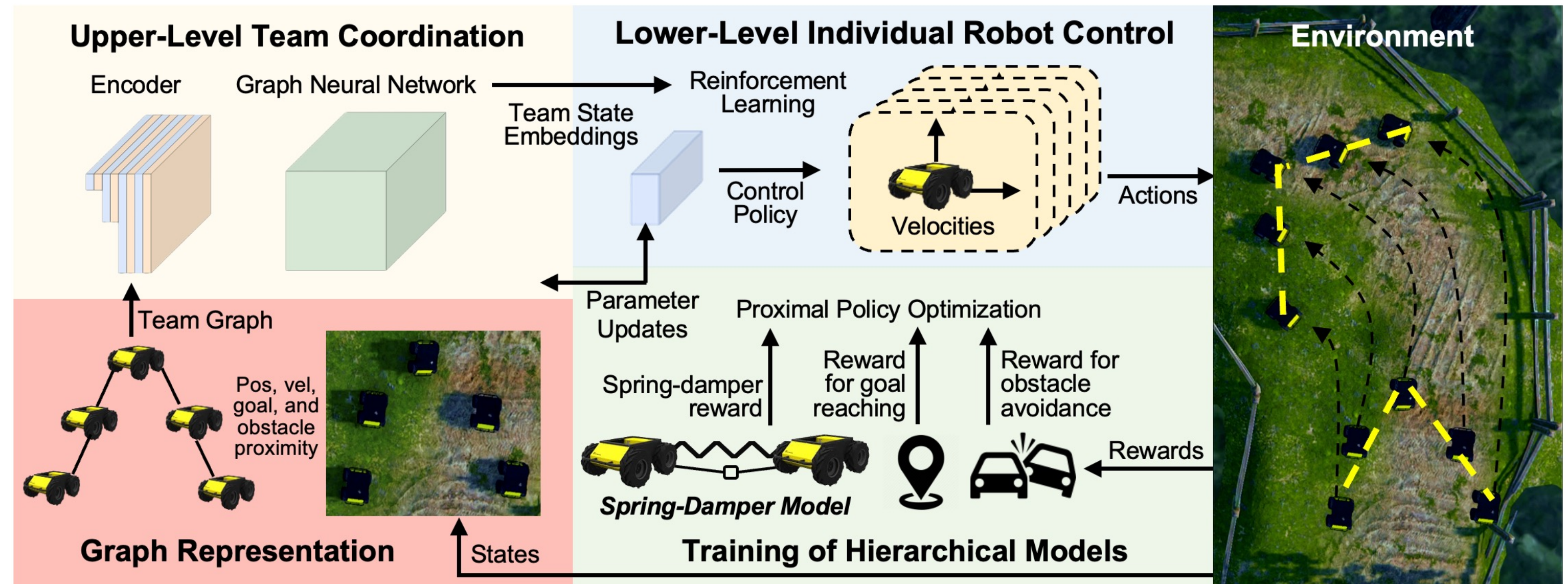
- Coordinated multi-robot navigation is an essential ability for a team of robots operating in diverse environments.
- Robot teams often need to maintain specific formations, such as wedge formations, to enhance visibility, positioning, and efficiency during fast movement.
- Real-world environments are often complex and include narrow corridors, where rigid team formations cannot maneuver effectively.
- We address the challenge of adaptive formation control in coordinated multi-robot navigation, which enables robots to dynamically adapt their formations for navigating narrow corridors, while moving more stably towards their goal.



Approach

- We represent a team of robots as a graph and propose *Adaptive Formation with Oscillation Reduction* (AFOR) with a hierarchical learning formulation.
- At the upper level, GNN is used to facilitate information sharing and generate team state embeddings h_i based on individual state embeddings z_i , which enables team decisions for robot coordination and formation control.

$$h_i = W^h z_i + \sum_{j \in \mathcal{N}(i)} W^h (z_j - z_i)$$



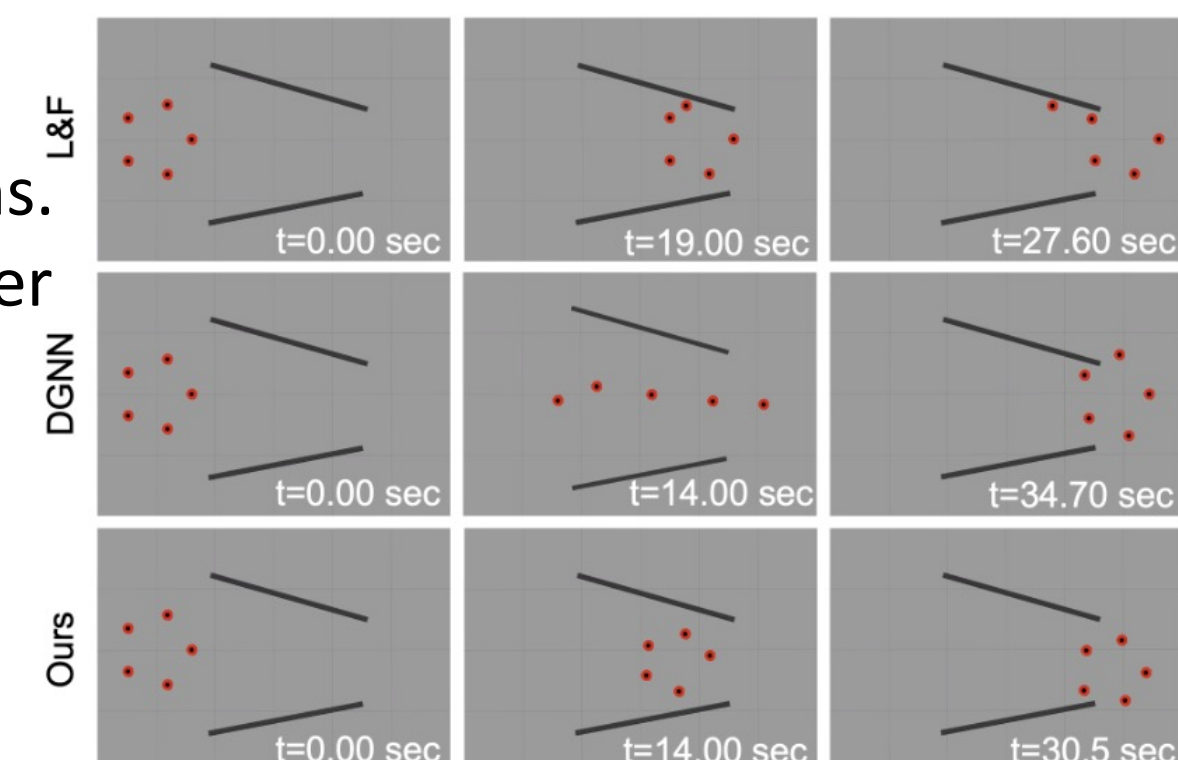
- At the lower level, given the h_i , a network ϕ is used to generate actions $a_i = \phi(h_i)$. RL is used to govern the low-level control, and learn policy π of individual robot control for navigation and obstacle avoidance.
- A spring-damper model is integrated with the hierarchical learning into AFOR to simulate flexible connections between robots, which is incorporated into the reward function to ensure adaptive formation and smooth navigation.
- PPO is used to train AFOR as a hierarchical model, which simultaneously updates weights W^h of GNN to generate team-level embeddings h_i at the upper level to facilitate team coordination, and the policy π at the lower level for individual robot control.

Experiments

Results on Formation Adaptation in Gazebo Simulations

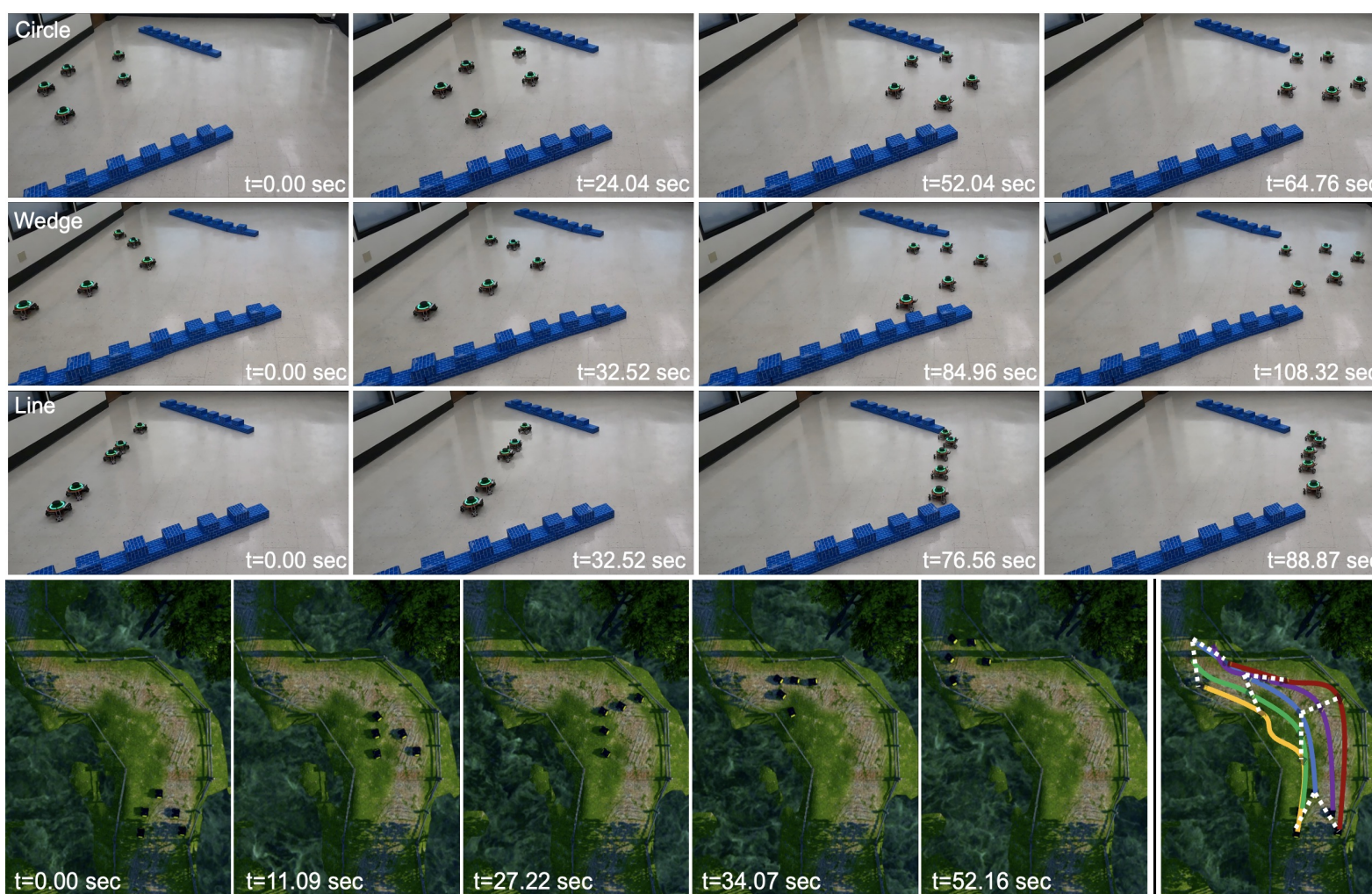
- Navigation capabilities of three formations are tested through a narrow corridors in Gazebo simulations.
- Evaluation metrics include successful rate and contextual formation integrity, smaller δ indicates stricter formation requirements.

Method	Circle Formation					Wedge Formation					Line Formation				
	SR (%)	$\delta < 0.5$	$\delta < 0.3$	$\delta < 0.1$	$\delta < 0.03$	SR (%)	$\delta < 0.5$	$\delta < 0.3$	$\delta < 0.1$	$\delta < 0.03$	SR (%)	$\delta < 0.5$	$\delta < 0.3$	$\delta < 0.1$	$\delta < 0.03$
L&F [10]	80	75.114	69.107	70.105	66.043	60	81.436	68.335	66.691	62.864	60	63.758	59.861	55.588	55.588
DGNN [46]	100	60.413	59.330	58.906	58.906	60	47.850	44.935	42.325	41.917	100	27.895	22.122	20.155	20.155
Baseline	100	84.450	84.097	80.521	79.880	100	86.506	76.491	68.219	66.733	100	82.918	81.731	64.264	64.438
AFOR (ours)	100	92.887	90.891	90.397	88.662	100	91.391	91.841	90.354	87.959	100	88.596	84.525	85.131	72.901



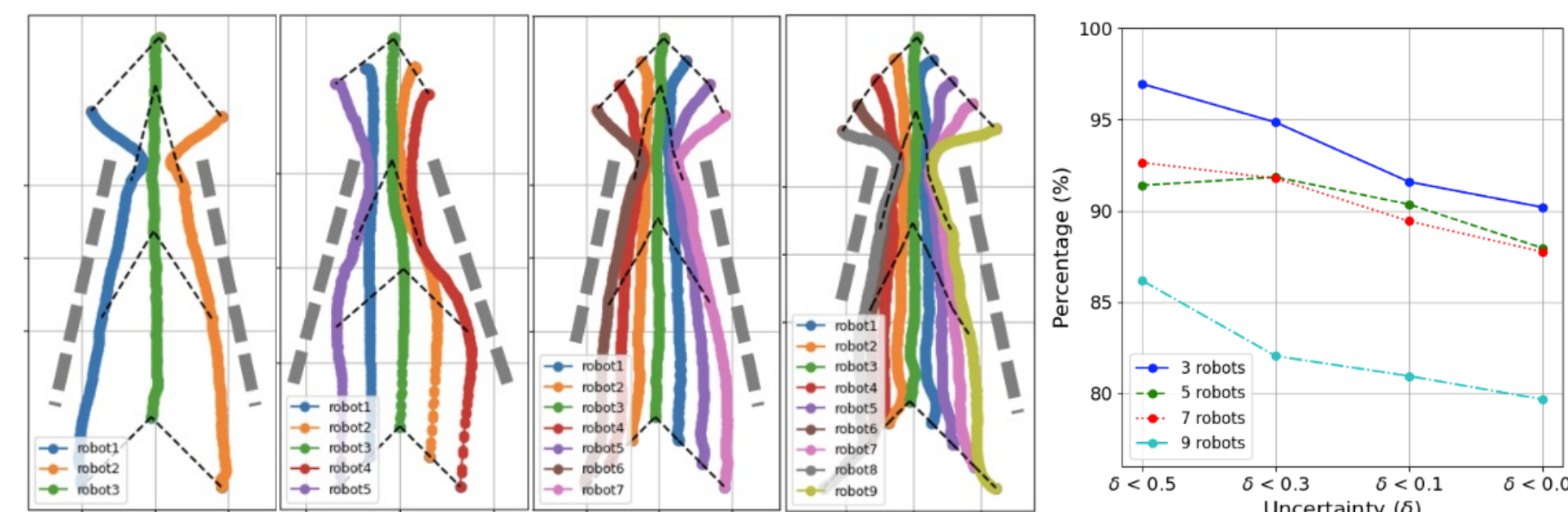
Case Studies on Real Robots and High-fidelity Simulations

- Five holonomic wheeled robots are deployed in a real-world environment. Ground truth positions are provided from OptiTrack system.
- Five differential-drive Husky robots are deployed in a realistic and challenging outdoor environment.



Discussion on AFOR's Generalizability

- AFOR is evaluated by different number of robots, and the results show their trajectories along with contextual formation integrity.



Summary

- This work introduces one of the first learning solutions for formation adaptation with oscillation reduction.
- We introduce a novel hierarchical learning method that incorporates a spring-damper model to simultaneously learn team decision-making for coordination at the upper level, and individual robot control at the lower level.

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Contact

Zihao Deng
Human-Centered
Robotics Lab
CICS @ UMass Amherst
zihao.deng@umass.edu



hcrclab.gitlab.io/project/afor

